

CS & IT ENGINEERING

Programming in C

Operators :02

Lecture No: 05



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TOPICS TO BE COVERED



Operators

02 Discussion

Q.1



What is the output of the given c program

```
void main()
```

```
{
```

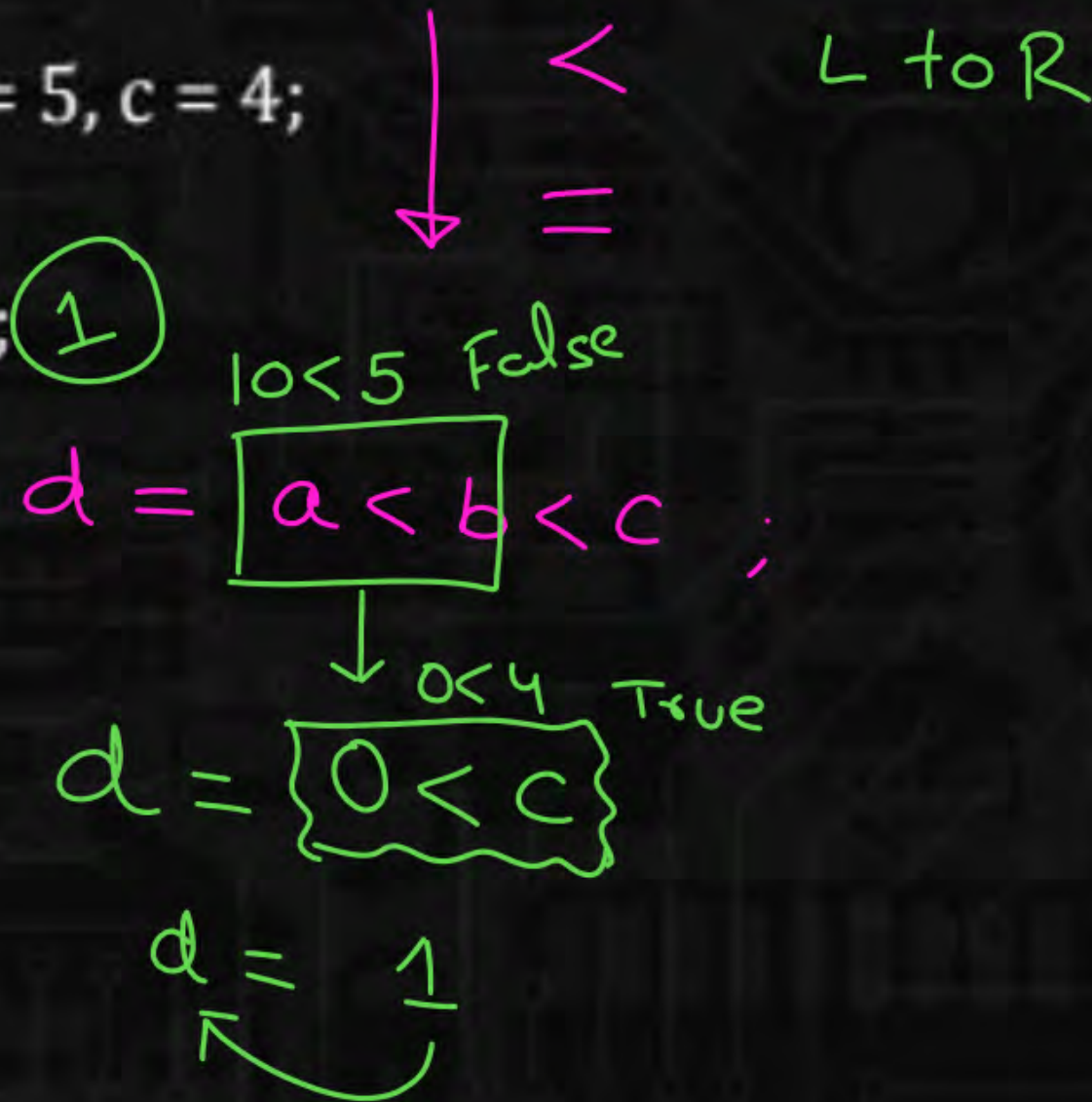
```
int d, a = 10, b = 5, c = 4;
```

```
d = a < b < c;
```

```
printf("%d", d);
```

```
getch();
```

```
}
```

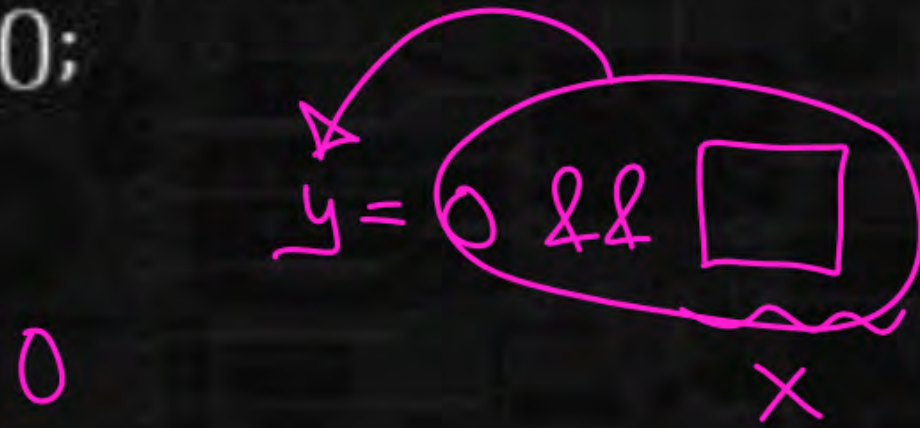


Q.2



What is the output of the given C program.

```
void main()  
{  
    int x = 0, y;  
    y = x && printf("PankajSir");  
    printf(" %d", y);  
    getch();  
}
```



Q.3

What is the output of the given C program.

```
void main()
{
    int x = 5, y;
    y = x && printf("PankajSir");
    printf("\n%d", y);
    printf("PankajSir");
}
```

PankajSir
1

PankajSir 1 X

PankajSir
1

5 && ✓ printf("PankajSir")

y = 5 && 9
 ↓ ↓
non-zero non-zero
[y = 1]

Q.4



What is the output of the following given C program

```
void main()
```

```
{
```

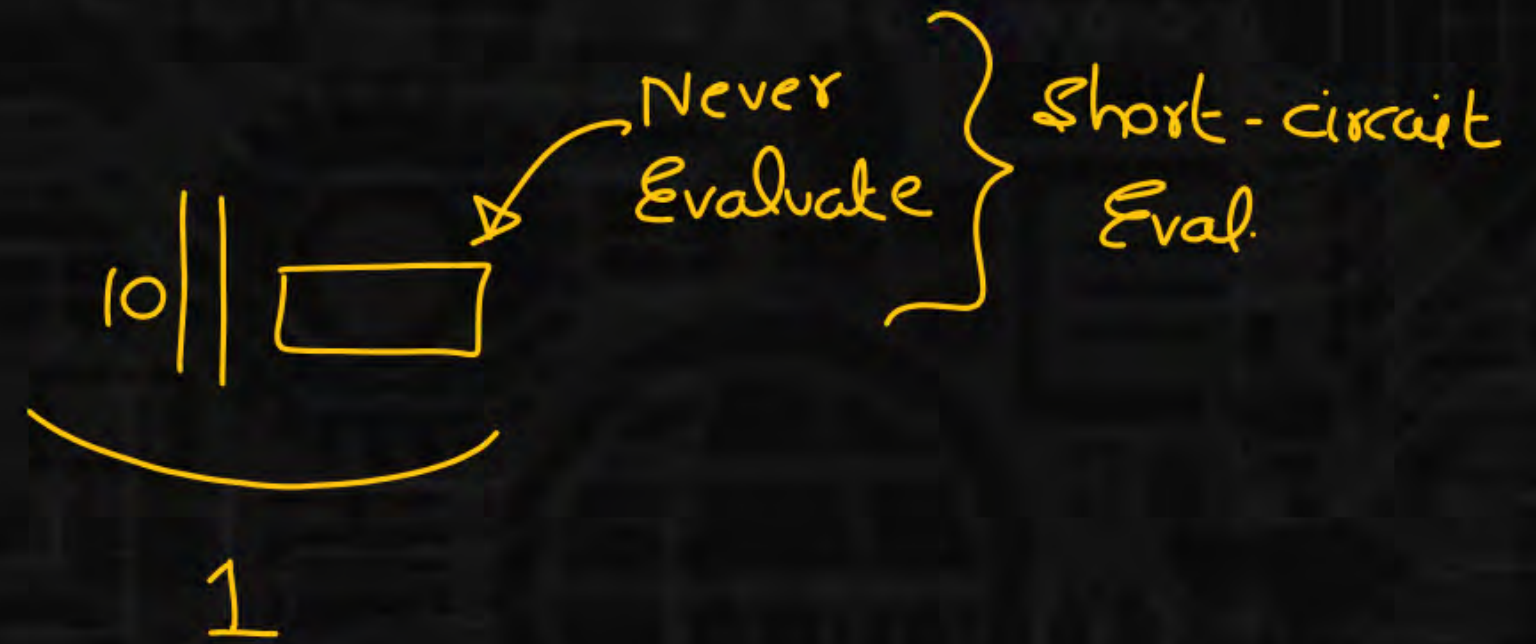
```
    int x = 10, y;
```

```
    y = x || printf("students");
```

```
    printf("\n%d", y);
```

```
    getch();
```

```
}
```



A.

Students10

B.

students

10

C.

10

D.

10

students

~~(E)~~ None of these

More assignment operators

$$x = x + 2 \quad \Rightarrow \quad x += 2 \quad \quad + =$$

$$x = x - 2 \quad \Rightarrow \quad x -= 2 \quad \quad - =$$

$$x = x \times 2 \quad \Rightarrow \quad x \times = 2 \quad \quad \times =$$

$$x = x / 2 \quad \Rightarrow \quad x /= 2 \quad \quad / =$$

$$x = x \cdot / 2 \quad \Rightarrow \quad x \cdot /= 2 \quad \quad \cdot / =$$

$$x = x \ll 2 \quad \Rightarrow \quad x \ll = 2 \quad \quad \ll =$$

$$x = x \gg 2 \quad \Rightarrow \quad x \gg = 2 \quad \quad \gg =$$

$$x = x \& 2 \quad \Rightarrow \quad x \& = 2 \quad \quad \& =$$

$$x = x | 2 \quad \Rightarrow \quad x | = 2 \quad \quad | =$$

$$x = x ^ 2 \quad \Rightarrow \quad x ^ = 2 \quad \quad ^ =$$

Q.5



Consider the following C program.

```
void main()
```

```
{
```

```
    int x = -5;
```

```
    int p, q;
```

```
    p = ++x;
```

```
    q = x++;
```

```
    --p;
```

```
    q++;
```

```
    p += q;
```

```
    printf("%d", p);
```

```
    return 0;
```

```
}
```

A.

10

B.

11

C.

12

D.

13

(E) None of these

-8

The output of the following code is

Q.6



What is the output of the given C program.

```
void main()
```

```
{
```

```
    int x;
```

```
    x = 6 > 3 != 0;
```

```
    printf("%d", x);
```

```
}
```

$$x = 6 > 3 \neq 0 ;$$

$$x = 1 \neq 0$$

$$x = 1$$

1



Q.7



Find the output of the following C – Program

```
void main()
```

```
{
```

```
    int x;
```

```
    x = 5! = 2! = 9 == 1 < 5! = 0;
```

```
    printf("%d", x);
```

```
    getch();
```

```
}
```

$x = 5! = 2! = 9 == 1 < 5! = 0;$ (True)

$x = 5! = 2! = 9 == 1! = 0;$ (True)

$x = 1! = 9 == 1! = 0;$

$x = 1 == 1! = 0$

$x = 1! = 0$

$x = 1$

$\left. \begin{matrix} < \\ !=, == \end{matrix} \right\} \text{L to R}$

Q.8

Consider the following C – program.

```
void main()
{
    int a;
    a = !!! Printf("PankajSir");
    printf("\n%d", a);
    getch();
}
```

A. PankajSir

B.

1

PankajSir

C.

PankajSir

1

☒ D.

PankajSir

0

! $\Rightarrow$ unary $\} R \text{ to } L$

!!! Printf("PankajSir")

!(!(!9))

!(!0)

!1

$\Rightarrow$ 0

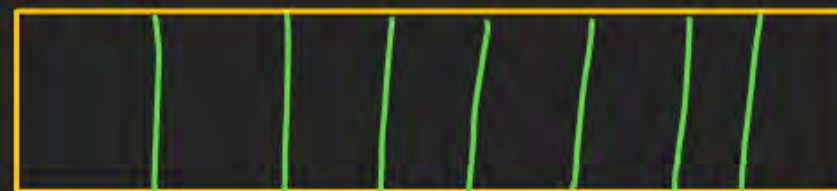
PankajSir -
0

Bitwise NOT (~)

Signed Magnitude :

$+8 \Rightarrow 0$
 $-8 \Rightarrow 1$

MSB



Sign
bit

7 bits

$+8 \Rightarrow$ Sign-bit 0 (+ve)

Magnitude $\Rightarrow 8 \Rightarrow 1000$
 $\Rightarrow 0001000$

$+8 \Rightarrow$



$$\left. \begin{array}{l} 0\ 0000000 \Rightarrow +0 \\ 1\ 0000000 \Rightarrow -0 \end{array} \right\}$$

2's complementation

+ve numbers $\Rightarrow$ as it is (binary form)

-ve numbers $\Rightarrow$ 2's complementation form

8 bits

+11 $\Rightarrow$

2	11	Rem
2	5	1
2	2	1
2	1	0
	0	1

$\uparrow$

1011

+11 $\Rightarrow$ 00001011

-11 $\Rightarrow$ (i) Find +11 binary

(ii) Find its 2's complement

-11 $\Rightarrow$ (i) +11 $\Rightarrow$ 00001011

(ii) 2's complementation

a) Find 1's comp.

b) Add 1 to 1's comp

direct
method

1's comp $\Rightarrow$ Flip all bits

0 $\rightarrow$ 1

1 $\rightarrow$ 0

$$-11 \Rightarrow (i) + 11 \Rightarrow 00001011$$

$$1s\ comp \Rightarrow 11110100$$

+1

2's comp.

$$\begin{array}{r} 11110101 \\ \hline \end{array}$$

Focus on 0's

$$\begin{array}{ccccccc} 1 & 1 & 1 & 1 & 0 & 1 & 0 & 1 \\ 2^7 & 2^6 & 2^5 & 2^4 & 2^3 & 2^2 & 2^1 & 2^0 \end{array}$$

$$\begin{aligned} &= -2^3 - 2^1 - 1 \\ &= -8 - 2 - 1 = \boxed{-11} \end{aligned}$$

$$0+1=1$$

$$1+0=1$$

$$0+0=0$$

$$1+1 \Rightarrow \boxed{10} \text{ } 2$$

$$9+1 \Rightarrow 10 \text{ } \swarrow \text{Carry}$$

$$\begin{array}{c} \swarrow \text{Carry} \\ 1+1=10 \\ \downarrow \text{Sum} \end{array}$$

$$\begin{array}{c} 1+1=10 \\ \downarrow \text{Sum} \end{array}$$

$$\begin{array}{ccccccc} \textcircled{1} & 1 & 1 & 1 & 0 & 0 & 0 & 1 \\ & & & & 2^3 & 2^2 & 2^1 & 2^0 \end{array}$$

$\Rightarrow ?$

$$-2^3 - 2^2 - 2^1 - 1$$

$$= -8 - 4 - 2 - 1$$

$$\Rightarrow \textcircled{-15}$$

$\sim$ $\Rightarrow$ is complementation

int a = 48, b;

b = $\sim$ a;

printf("/d", b);

$$a \Rightarrow (2^5 + 2^4)$$

6 5 4 3 2 1 0
2 2 2 2 2 2 2

48 : 00000000000000110000

(b) ~ 48 : 1111111111001111

2^1 s comp.

$$-2^5 - 2^4 - 1$$

$$= -(2^5 + 2^4 + 1)$$

$$\sim a = -(a+1) \checkmark$$

$$\neg a \Rightarrow -(a+1)$$

```
int a = -5;
```

```
printf("%d", 4 \neg a);
```

$$-(a+1)$$

$$-(-5+1)$$

$$-(-4) \Rightarrow 4$$

Ternary Operator (?)

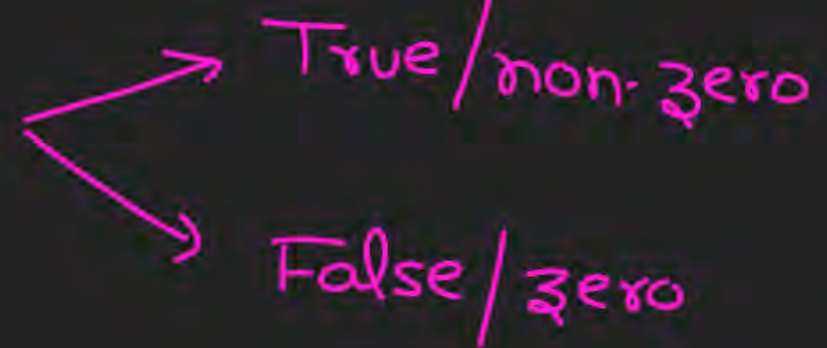
Syntax:

$\text{exp1} \text{ ? } \text{exp2} : \text{exp3}$



The diagram shows the syntax $\text{exp1} \text{ ? } \text{exp2} : \text{exp3}$ with three double-headed arrows underneath. The first arrow is under exp1 , the second is under exp2 , and the third is under exp3 .

(i) exp1 is evaluated



The diagram shows a branching structure where the text "(i) exp1 is evaluated" has two arrows pointing to "True/non-zero" and "False/zero".

- True/non-zero
- False/zero

If exp1 is True(non-zero) then result $\Rightarrow \text{exp2}$
If exp1 is False(zero) then result $\Rightarrow \text{exp3}$

```
int max;  
max = 10 > 7 ? 10 : 7 ;
```

True

```
a = printf("2");  
a = 1
```

max = 10

1 int a;

O/P: 121

```
a = printf("1") ? printf("2") : printf("3");
```

← exp1 → ← exp2 → ← exp3 →

```
printf("/d", a);
```

1 ? printf("2") : printf("3");

True

Q.9

Consider the following C – program.

```
void main( )
{
    int x;
    x = (~ 5 | 2) > ~ 0 ? 9 : 10 << 2;
    printf("%d", x);
}
```

The output of the following code is 40.

$$\begin{aligned}x &= 10 \ll 2 \\&= 10 \times 2^2 \\&= 10 \times 4 \\&= 40\end{aligned}$$

$$\begin{aligned}\sim 0 &= -(0+1) = -1 \\ \sim 5 &= -(5+1) = -6\end{aligned}$$



$$(\underbrace{\sim 5 | 2}_{\text{exp}_1}) > \underbrace{\sim 0}_{\text{exp}_2} ? 9 : \underbrace{10 \ll 2}_{\text{exp}_3}$$

$$(-6 | 2) > \sim 0$$

$$\begin{array}{r}11111010 \\ 00000010 \\ \hline 11111010 \\ -6 > -1 \text{ False}\end{array}$$

Q.10

Consider the following C – program.

```
void main()
```

```
{
```

```
     $x=5,$ 
```

```
    int  $y=10, z, p, q;$ 
```

```
     $z = ((x \leq y) \parallel (x \neq y));$ 
```

```
     $p = ((x < y) \&\& (x == y));$ 
```

```
     $q = !! ((x < y) \&\& (x \neq y));$ 
```

```
    printf("%d", q);
```

```
    return 0;
```

```
}
```

The output of the following code is ____.

 z $(5 \leq 10) \parallel (x \neq y)$ $z = 1$ $p = 0$ $q = !! 1$ $= 10$ $= 1$ $q = 10 \times z + p - 2 \times q$ $q = 10 \times 1 + 0$ $- 2 \times 1$ $= 10 - 2$ $= 8$ 

Q.11



What would be the output of the following C – program given below?

```
int main()  
{  
    int x, y, z;  
    x = 6 < 5 > 2;  
    y = 8 > 5 < 0;  
    z = 5 > 4 != 0;  
    printf("%d%d%d", x, y, z);  
    return 0;  
}
```

A.

0 1 0

B.

1 1 1

C.

0 0 1

D.

1 0 1

Q.12



What would be the output of given C – program?

```
int main()
{
    int x, y, z;
    x = 5, y = 10;
    z = x == y;
    printf("%d", z);
    return 0;
}
```

A.

0

B.

1

C.

Garbage value

D.

Compiler error

Q.13

Consider the following C – program

```
int main()  
{  
    int x = -9, y = 7, z, p, q;  
    z = x % y;  
    p = (-x) % (-y);  
    q = z + p;  
    printf("%d", a);  
    return 0;  
}
```



Q.14



Consider the following C – program

```
int main()  
{  
    int x , y;  
    x = 20, 30, 40;  
    y = (20, 30, 40);  
    printf(“%d%d”, x, y);  
    return 0;  
}
```

The output of the given code is

A. 20, 20

B. 20, 40

C. 40, 40

D. 40, 20

Q.15

What will be the output of the given C – code?

```
int main()
{
    int x = 12;
    (x & 1)? printf("GATE") : printf("2023");
    return 0;
}
```

2023

$$\begin{array}{r} 1100 \\ 0001 \\ \hline 0000 \\ \hline 0 \end{array}$$

`printf("%.d", 1+2x3-4);`
└─ Eval.

1+6-4
7-4
3

① `printf("%.d", printf("%.d", 1+2x3-4));`
└─ 1 symbol
└─ 31
└─ `printf("%.d", 1);`

```
int a;
```

```
a = printf("/d", printf("GATE")① == printf("2023"));  
printf("/d", a);
```

```
a = printf("/d", 4 == 4)1
```

| Symbol

GATE202311

a = 1

sizeof

* unary operator

* Compile time

$\text{sizeof}(-4) \Rightarrow 4$

$\text{sizeof}(2) \Rightarrow 4$

$\text{sizeof}(3+4) \Rightarrow 4$

$\left. \begin{array}{l} \text{ } \\ \text{ } \end{array} \right\} \text{int + int} \\ = \text{int}$

unsigned int

short - 2

int - 4

float - 4

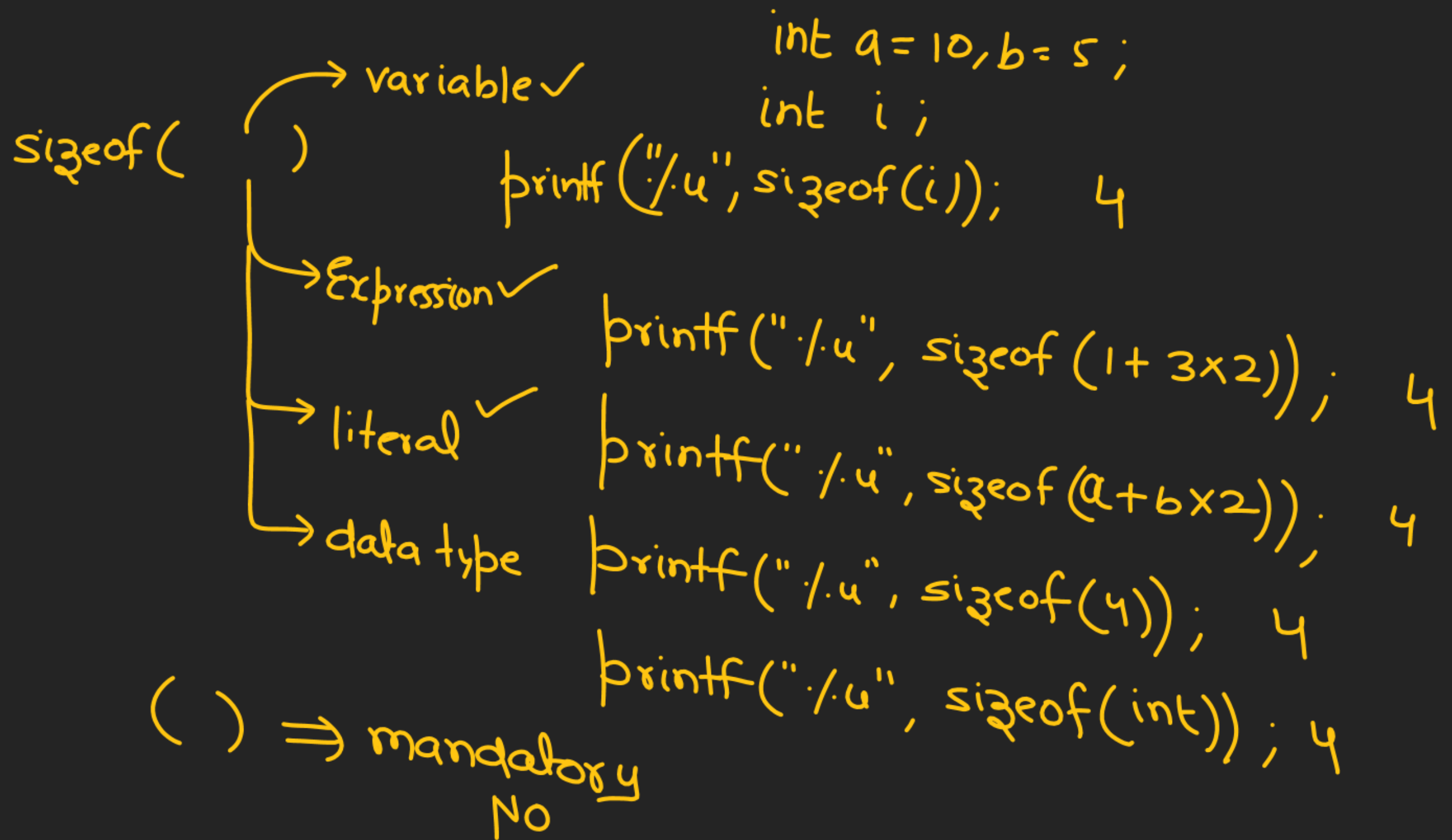
char - 1

$\text{sizeof}(\text{short}) \Rightarrow 2$

$\text{sizeof}(\text{char}) \Rightarrow 1$

$\text{sizeof}(\text{int}) \Rightarrow 4$

$\Rightarrow \text{+ve}$



<code>sizeof(4)</code>	or	<code>sizeof 4</code>	
<code>sizeof 4+3x2</code>		<code>sizeof(4+3x2)</code>	
<code>sizeof(int)</code>		<code>sizeof int</code>	X

`Char a,b;`
`sizeof(a+b)`
↓
~~~~~

Q.16

Which of the following is/are the bitwise operator?



A.

&

B.

&&

C.

<<

D.

|

Q.17



The output of the following C – program.

```
int main()  
{  
    int x = 5, 6, 7;  
    printf("%d", x);  
    return 0;  
}
```

A.

5

B.

7

C.

Garbage value

D.

Compiler error

Q.18



Consider the following C – program

```
int main()  
{  
    int x = 15;  
    printf("%d", (++ x)++);  
    return 0;  
}
```

The output of the program is \_\_\_\_.

A. 15

B. 16

C. 17

D. Compiler error

Q.19



Consider the following C – Program.

```
int main()  
{  
    int x = 10, y = 20, z = 30;  
    printf("%d", sizeof(z/= x + y));  
    printf("%d", z);  
    return 0;  
}
```

Assume size of an integer as 4 bytes. What is the output of above program?

A. 4 1

B. 4 3 0

C. 2 1

D. Compiler error

**Q.20**

What is the output of following program?

```
int main()  
{  
    int p = 1;  
    int q = 1;  
    int r = p || --q;  
    int s = p -- && --q;  
    int t = p + q + r + s;  
    printf("%d", t);  
    return 0;  
}
```

A.

1

B.

2

C.

0

D.

4

**Q.21**

What would be the output of the following C – program?

```
int main()  
{  
    int x = 2, y = 5, z;  
    x = x^y;  
    y = y^x;  
    z = x - y;  
    printf("%d", z);  
    return 0;  
}
```

**Q.22**

The output of the given C – program.

```
int main()  
{  
    int x = 0;  
    int y = 0;  
    x = (x == (x == 1));  
    printf("%d", z);  
    return 0;  
}
```

**Q.23**

What will be the output of the given C – program

```
int main()
{
    int a = 0, b = 1, c = -1, x = 0.5, y = 0.0, z;
    z = x*y < a + b || c;
    printf("%d", z);
    return 0;
}
```

A.

1

B.

2

C.

-1

D.

0

**Q.24**

What will be the output of the given C – program

```
int main()  
{  
    int x;  
    x = 3 == 3 > = 5 != 0 == 0;  
    printf("%d", x);  
    return 0;  
}
```

**Q.25**

Consider the following code snippet.

```
int main()  
{  
    int x;  
    x = printf("%d", printf("GATE2023"));  
    printf("%d", x);  
    return 0;  
}
```

The output of the given code is \_\_.

```
int a=0, b=2, c=0, d=3, e;
```

```
e = ++a || b-- && c++ && --d
```

```
printf("/.d/.d/.d/.d/.d", a, b, c, d, e);
```

2:00 PM

